

Design and Implementation of an Intelligent Bird Feeder for Wildlife Monitoring

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Abstract— This paper presents the design and implementation of the prototype of a smart bird feeder, incorporating several sensors for automated functions. The article outlines the components and rationale behind the design and use of all parts of the prototype, as well as the coding logic for the sensors. The system integrates 4 sensors and 3 outputs to track bird presence and feeder interactions. It also includes a list of improvements that could be incorporated into future iterations of the prototype, both in the form of hardware and software. The experimental results show reliable distance measurements, weight detection, light adaptation and responsive motor reactions, ultimately confirming the successful realization of a smart bird feeder and monitor.

Index Terms— Arduino, Birds, Electronics, Embedded systems, Feeder, Sensors

I. INTRODUCTION

In the context of the ELG 3336 course, we are tasked with the design of a system containing at least 3 sensors. For the system, the team decided to create a bird feeder with the following objectives:

1. Feed various species of birds [1].
2. Prevent squirrels from accessing the food.
3. Dispense a controlled amount of food for each bird appearance.
4. Take pictures of the birds to collect data (species, weight, frequency of visits)

Using an Arduino ESP32 as the microcontroller, the sensors featured on the final feeder prototype are:

1. Photocell [7].
2. Weight sensor [8].
3. Ultrasonic sensor [9].
4. Camera.

The feeder will therefore dispense a certain amount of seeds (or any other food of the user's choosing) by meeting 2 requirements:

1. If a bird is within range, determined by the ultrasonic sensor.
2. If the weight sensor provides a valid reading,

There is also a threshold coded into the weight sensor, for which the value of the threshold is lower than the weight of an average squirrel. A typical squirrel on the feeder would therefore invalidate the feeding mechanism to trigger.

Finally, the photocell works in conjunction with the camera, which takes a picture when a bird is detected for wildlife data tracking purposes. The photocell activates an LED [6] which serves to illuminate the photo, in case the ambient lighting is too low.

All these sensors work in conjunction with their code to create an all-weather, self-sustaining monitoring and data collection tool for researchers, while being non-intrusive.

II. LITERATURE REVIEW

A. Minimizing false positives in electronics: Debouncing

The idea behind a dual condition requirement for the activation of the dispensing system is because, in our case, the ultrasonic sensor alone can be triggered by a multitude of events that are not due to the desired triggers. For instance, instead of a bird landing within the set distance of the sensor, the sensor could very well trigger due to a falling leaf, rain, another animal, or even a curious human. Therefore, the system is designed with an auxiliary sensor to confirm the presence of our target, which is small birds. The metric chosen to narrow down the potential triggers is the weight exerted onto a loading cell placed in front of the feeding area. The trigger object now needs to be within the distance and weight thresholds to activate the mechanism.

Most backyard feeder birds in North America weigh no more than 30 grams [1], with the blue jay being on the bigger and thus heavier side, ranging from 70-100 grams [1]. On the other hand, squirrel species weigh much more, with the smallest species being the red squirrel, coming in at 200 grams at their smallest [2], while larger species, such as the grey squirrel, weigh more than 500 grams [3]. Based on these values, the value for the threshold should be somewhere between 100 grams and 200 grams, staying at around 150 grams to give a margin for outliers, such as big birds or small squirrels.

When using a weight sensor with a high sampling rate, a heavier object can bypass the threshold condition if the weight is loaded slowly. In other words, in the intermittent step that a squirrel makes to climb onto the feeder by only putting its front legs onto the load cell, a reading lower than the set weight threshold can be read by the sensor because of its high sampling rate. A simple solution is to reduce the sample rate in the range of 1 sample per second, but it is not an elegant solution, since it is possible that the situation described above with the squirrel happens due to unfortunate

timing. In other words, it is a possibility that the squirrel loads part of its bodyweight *right* when the sensor samples a reading, falsely resulting in a small weight reading. A more reliable method is to include a debounce in the code, which is a time-based validation [4]. The working principle is the following: once a weight below the threshold is detected, the code checks for the weight again after a set time delay. If the weight remains under the allowed limit, then the feeder is allowed to activate. Of course, if the weight is initially above the threshold, the code should deny the feeding mechanism from starting.

B. Finite state machines and fail safes

While the dual-sensor validation reduces false positives, the system must also consider scenarios where activation or denial could leave it stuck in an unintended state. For instance, if a bird remains on the feeder longer than intended, or if a heavy object is detected, the system could remain in a locked-out state indefinitely. An effective solution is to implement a time-based auto-reset function [5] to the feeding code logic. In essence, after a decision to open or keep the feeder closed is made, the system resets to its initial condition after a timer has elapsed, ensuring uninterrupted operation.

C. Visual monitoring, day and night

Light-emitting diodes (LEDs) are known for their very efficient power consumption to achieve light output [6], and therefore they are a perfect choice for our application which is lighting up the camera frame when dark. Most wildlife tracking cameras make use of infrared (IR) lights and IR cameras as to not disturb or interfere with wildlife. Unfortunately, our available components constrained the use of LEDs to the human visual spectrum. The main advantage of an IR system is that birds probably will not be scared off because they do not emit a sudden burst of bright light unlike normal LEDs, resulting in more consistent feeding sessions and more accurate weight data.

D. Working principles of each sensor

The ultrasonic determines the distance between it and any object by sending out a wave and capturing the same wave that reflects back [7]. The sensor assumes a constant speed of sound travelling in see level air as a basis for its calculations.

$$d = \frac{v \times t}{2} \quad (1)$$

Where d is the distance between the sensor and the surface of the detected object, v is the speed of sound, t is the elapsed time for the soundwave to travel from the sensor and back.

The load cell measures a load applied orthogonally to its surface. It uses strain gauges bonded to the metal surface of the device. When strain occurs, the elongated or shortened strain gauges change in resistance. The Wheatstone bridge layout of the gauges allows for a measurable output voltage, which is proportional to the applied load [8].

The photocell is essentially a resistor, but its resistance changes with the amount of light it receives. It uses

semiconductor photoconductivity, for which the energy of photons hit the free electrons in the semiconductors, causing a lower resistance [9].

III. INVESTIGATION DETAILS

A. System physical design

The feeder consists of laser-cut 1/8th inch medium-density fiberboard (MDF), easily assembled into the house shape without tools. The onboard sensors are secured via bolts, nuts and extra MDF acting as mounting brackets. All the wiring from the sensors route to the center of the feeder, which houses the Arduino Uno board. Also housed in the center of the feeder are the feeding assembly, consisting of a reservoir made from part of a plastic water bottle, and the dispensing unit, a servo actuated panel which blocks the dispensing of the seeds when the servo is in the “closed” position, or moves out of the way to open when instructed by the Arduino.

Complementary features include pre-cut holes on the side pieces of the structure, allowing the feeder to be hung from a tree, a pole or any elevated structure, and a counterweight, to keep the feeder level due to the asymmetrical mounting of the load cell on only one side of the structure.

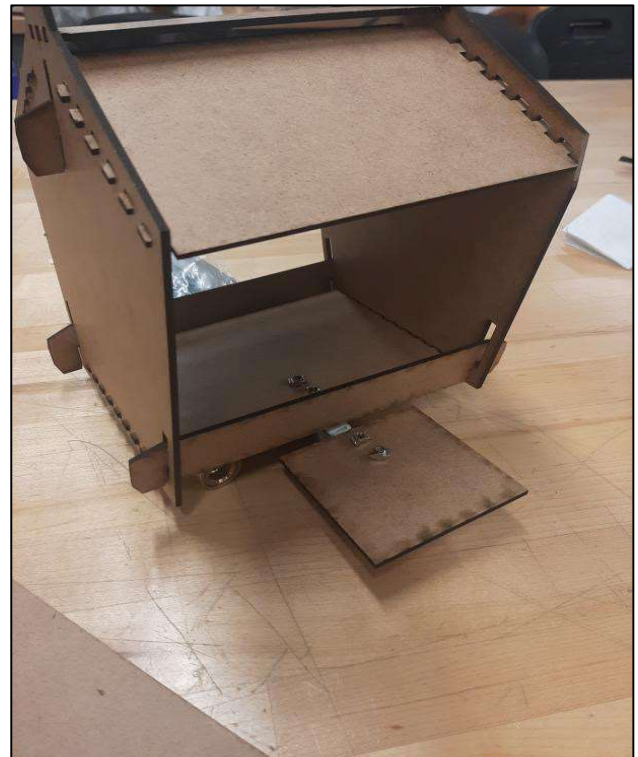


Fig. 1. MDF construction of the feeder body

B. Embedded code operation

The programming developed for our smart bird feeder coordinates all the sensors, the outputs and the data-capture functionalities using the ESP32 microcontroller. The program starts by including all the necessary libraries for the ESP32, the camera operation, the SD-card data storage, and the servo

controls. We then define all the GPIO pins to interface with the system's components, including one or two ultrasonic sensors, a light sensor and LED pair, the servo module, the weight sensor (after the first prototype) and the ESP32-compatible camera module. The code continues with a full camera configuration. That part of the code ensures that each camera signal such as data lines, clock pins, or synchronization pins, is related to the right ESP32's hardware pins. After this, the SD interface is created. If either the camera or SD card fails to initialize, the system stops to prevent any errors. Once the camera module is done, it's the servo motor's turn. The motor is configured to a control pin and initially set to the closed position. Ultrasonic sensors are also configured with dedicated trigger and echo pins. Then, the LED output pin is also configured to match the LDR input pin.

During the main loop, the program reads distance measurements from ultrasonic sensors and ambient light from the LDR. These values are printed on the serial monitor. When an object is detected within 15 cm, the system gets ready for feeding. There's a delay between feeding cycle to prevent overfeeding. The cycles is coded as such:

1. The servo motor opens
2. A photograph is taken and stored on the SD card.
3. The timestamp for the last feeding is updated.
4. A short delay at the end of each loop

The LED is automatically enabled when lighting falls below a threshold, thus the camera can capture clear images.

```

161 void loop() {
162
163     float dist1 = readDistanceCM(trig1, echo1);
164     float dist2 = readDistanceCM(trig2, echo2);
165     int lightValue = analogRead(ldrPin);
166
167     Serial.printf("Light: %d | D1: %.1f | D2: %.1f\n", lightValue, dist1, dist2);
168
169     // LIGHT AUTO-ON
170     if (lightValue < 1800) digitalWrite(ledPin, HIGH);
171     else digitalWrite(ledPin, LOW);
172
173     // BIRD DETECTION
174     if ((dist1 < 15 || dist2 < 15) && millis() - lastFeedTime > feedDelay) {
175         Serial.println("🐦 Bird Detected! Feeding + Photo...");
176         myServo.write(50);
177         delay(2000);
178         myServo.write(0);
179
180         capturePhoto();
181         lastFeedTime = millis();
182     }
183
184     delay(300);

```

Fig. 2. Loop Code of the sensors

C. System logic

The functioning of the system starts with sensors that record data. The sensors have their own amplifier and filters to carry the data to the multiplexer in our ESP32. The ESP32 then acts as the multiplexer, sample and hold, ADC and controller that

leads to data processing. Arduino IDE processes the data and sends a signal to the outputting elements such as the LED, the servo motor or the SD card.

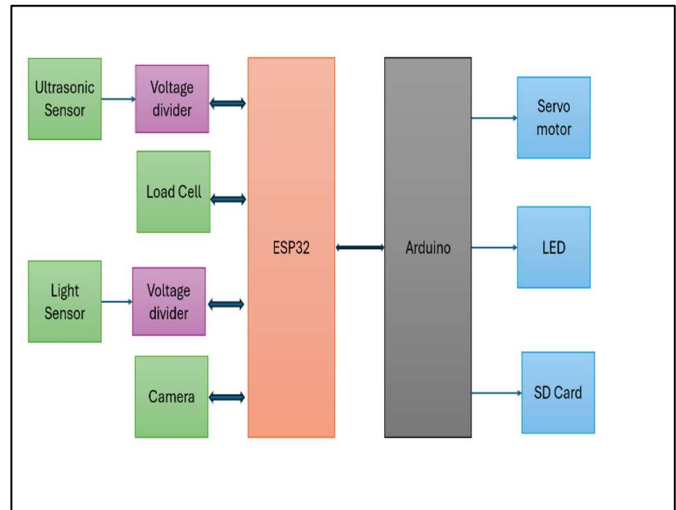


Fig. 3. Block Diagram of the System

D. Sensor Integration

The ESP32 implements an analog-to-digital converter (ADC) and so it measures voltage rather than resistance. For that reason, in the circuit some sensors had to use a couple elements studied in the course. For example, The LDR had to be integrated with a voltage divider composed resistor a 10 kΩ reference resistor. The two elements were arranged in series between 3.3 V and ground, with the LDR input connected to pin GPIO 34. The ultrasonic sensor also uses voltage division with two resistors on it's trigger connection. The sensor uses a digital trigger pin to emit a pulse and an echo pin to capture the return time of the reflected sound wave, as well as a connection to 5V and ground. The load cell stationed under the bird perch is also connected based on theory studied in experiment. More specifically, the sensor works as a Wheatstone bridge and an external amplifier module HX711 amplifies the output before the transmission to the ESP32. The servo motor, on the other hand, has the simplest connections, powered by 5V and connected to ground.

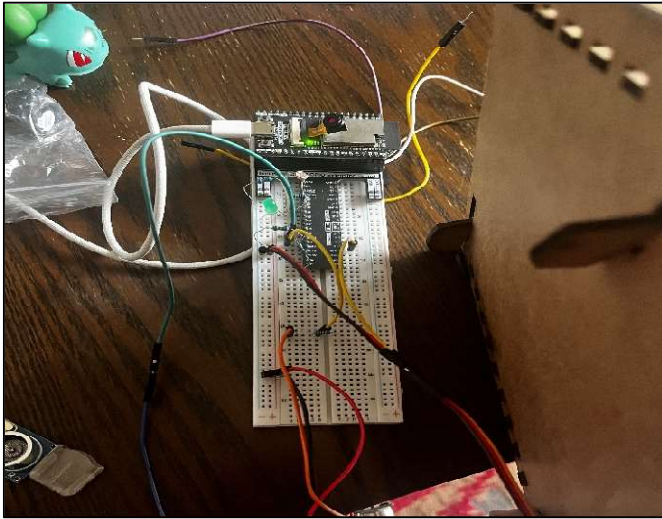


Fig. 4. Circuit connection with voltage dividers

IV. DISCUSSION

The Smart Bird Feeder is a good demonstration of embedded systems adapted for environmental monitoring. The combination of sensors such as light, motion, weight, and camera, allow automated monitoring of bird activity and feeder status. The system overview highlights the functioning of the project: each sensor performs a distinct function to help fulfill the goal of the feeder.

A. System Overview

For an architecture standpoint, the core of our smart bird feeder is composed of six main components. First, an ESP32 microcontroller serves as the central controller. It is especially useful for offering Wi-Fi connection, ADC channels, specific pins, and a camera module support. The camera module is the second element. It helps provide a visual verification for different situations and provides species identification potential. In third, we have the ultrasonic sensor. This sensor acts as one of the main components of the system as it enables measurement of bird distance to only open the feeder when needed. The ultrasonic sensors ultimately reduces mechanical wear of the feeder door opening as well as environmental interference in the whole feeder mechanism. At the heart of that mechanism, the servo motor acts as an electromechanical element, controlling the opening of the feeder door and allowing automated dispensing based on sensor inputs. The weight sensor, a load cell, brings a new capability: distinguishing species based on characteristic mass ranges. Finally, the light sensor and LED pairing reads environmental lighting and adapt illumination when brightness drops below a threshold. The actual body of our model is made of MDF, with a plastic feeder. Both materials made for simple, light and low-cost solutions for the prototype of the smart bird feeder; however, they are most likely going to be subject to replacement with more durable materials as the product gets optimized.

B. Results

When discussing the results, we found that most subsystems functioned reliably during testing. The ultrasonic sensor produced consistent distance readings, up to a distance 167 cm with minimal variance and accurate noise immunity. On the Arduino interface, it showed the distance of present wildlife before the feeding cycle enabled. Otherwise, it showed the distance to the nearest object. The camera module caused more issues. At first it didn't initialize and the manufacturer pins on the microcontroller didn't match those in the code. After we researched the correct connections and fixed the code, the module ultimately correctly synchronized with the ESP32, and we were able to capture images. The light sensor successfully turned the LED on, only under low-illumination conditions. In other words, the feeder would be visible thanks to the LED when the environment was dark, which confirms correct use of ADC thresholds. The load cell showed accurate readings but wasn't integrated easily with the system. To successfully add it to the sensors, the weight sensor needed calibration, then a specific reading condition in the programming of the software. Its definitive use was species-based classification and potential estimation of feed levels. The servo motor responded smoothly to command signals, opening and closing the feeder gate as expected.



Fig. 5. Bird Feeder prototype in action outside

Overall, the results reveal a system capable of monitoring birds and adapting. The weight sensor and camera together create a subsystem of species recognition that help with selective feeding and data collection. The ultrasonic sensor that enables the servo motor creates a second subsystem for a responsive mechanism in the feeder, that may be activated only when verified birds are present. It also helps reduce wasted feed and prevent unwanted wildlife access. Finally, the LED-assisted light sensor ensures that camera images stay usable in dim conditions, improving nighttime feed or detection performance.

C. Optimization and future iteration

Although the results are satisfactory, our project presents several limitations that must be acknowledged. In real

scenarios, the lighting variations of the outdoors may require calibration for both the camera exposure and the LDR sensor. For ultrasonic sensors and load cell, wind, rain, or other natural occurrences can create noise in readings. Both sensors would require digital filtering and more complex programming. On another hand, to keep up with power consumption, we also imagine an energy-efficient mode of charging, like solar charging for long-term placement of the feeder. Not only that, but mechanical parts, particularly the servo door mechanism, will also require weatherproofing to be able to resist humidity, dust, and temperature changes.



Fig. 6. Mechanical prototype of the Feeder system

An interesting feature to add would be the implementation of a species recognition software into the system, which could form a sorted list of pictures based on what the algorithm identifies along with the aid of the weight data from the load cell. Of course, a human can always double-check to confirm the correct placement of the birds in each category before finalizing the list.

Despite all, the system prototype lays a strong experimental foundation, confirming that a fully autonomous bird feeder is plausible. Like any good prototype, the project contains many future iterations to optimize the feeder. Future extensions comport cloud data logging, predictive feeding algorithms with a mobile app, or a more complex mechanical door mechanism. With refinement, we believe the device can evolve from a course project into a wildlife monitoring tool suitable for research or conservation purposes.

V. CONCLUSION

In conclusion, this article presented the design, implementation, and evaluation of bird feeder developed using an ESP32 microcontroller and multiple sensors. The goal of the project was to create a mechatronic system that could serve a purpose. We used the opportunity to make a feeder that would help collect data on biodiversity and wildlife, ultimately leaving

a better impact. Building on concepts from the course, we put in practice a system that combines a camera module, an ultrasonic distance sensor, a light sensor, a load-cell weight sensor and a servo- motor. These elements were selected and configured in alignment with principles discussed in the course notes about sensors and circuit building.

The research conducted for this project informed both the hardware selection and the system's operational logic. Articles on ultrasonic sensors trigger tendencies helped the design to be more aware of false positives by adopting safety measure with dual sensing, while results on loads cells motivated the use of time-based feedings. The findings on PWM control contributed to better understanding the servo movement, while finite-state machine principles helped us add an automatic reset to prevent the system from being stuck. Ultimately, the research led to better design processes for the firmware, the hardware and the mechanical elements.

Testing, Experimenting and Iterating our prototype demonstrated that our design was effective as a smart bird feeder. Although some sensors proved more difficult to implement, the core of the system demonstrated reliable behaviour. Indeed, the ultrasonic sensor had stable readings, the camera module captured images, the light sensor performed under different conditions and activated the LED when necessary. The servo motor also performed as expected with smooth and responsive reactions for the feeding mechanism. The load sensor has yet to perform but its implementation has been completed. Together, the sensor's results confirm that the system meets its core functional goals.

The feeder has a lot of optimization potential that could make it a real technological innovation. Future plans include new algorithms for species identification, environmental sensors, or adding solar-powered energy for durability. Incorporating these different elements would elevate the feeder to a more sophisticated ecological observation platform.

Overall, this project successfully demonstrates how research and multisensory designs can be used to create an intelligent bird feeder that monitors wildlife. Our prototype set a good base for continued development and meaningful contributions

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